Horizontal launches

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Learning target: Find flight time from vertical motion and then horizontal range.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 7.*
3 N/kg toward the center; divide the surface value by 25.

02. Worked example: explain the model
Height 20 m vertical; range 6 m horizontal. Determine fall time, then multiply horizontal velocity by time.

03. Read and interpret the model
Vertical motion determines time; labels must distinguish h from range.

04. Apply the physics
8 m; 12 m.

05. Explain your reasoning
No, it doubles time and range because fall time scales with√height.

06. Review: connect an earlier idea *Review: Unit 1, Day 20.*
$a=8\ m/s^2$ right; $v_f=40$ m/s right. Constant net force and mass throughout.

07. Exit check
15 m horizontally from base.

08. STRETCH: test the model
5 m/s hits target; 2.5 m/s lands 5 m from base.

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01. Warm-up: bring back an earlier idea *Review: Unit 2, Day 7.*
Center distances 3 R and 5 R; altitudes h/R=2 and 4.

02. Worked example: explain the model
Height 20 m vertical; range 6 m horizontal. Determine fall time, then multiply horizontal velocity by time.

03. Read and interpret the model
Vertical motion determines time; labels must distinguish h from range.

04. Apply the physics
t=√(90/10)=3 s; range 45 m.

05. Explain your reasoning
No, it doubles time and range because fall time scales with√height.

06. Review: connect an earlier idea *Review: Unit 1, Day 20.*
x interval [-66, -54] km; y interval [72, 88] km. Simplified geometry, gravity, Earth motion or missing observations limit the model. Bounds are not probabilities.

07. Exit check
t 2 s; range 16 m; vertical velocity 20 m/s down.

08. STRETCH: test the model
Time 1.5 s, height 11.25 m; vertical impact speed 15 m/s; total speed √369≈19.2 m/s.